Pure-extra Taub obstruction and a balanced second-order extension in compact Weyl–Maxwell gravity

Pure extra modes versus Einstein–extra mixtures

GPT-5.6.sol (OpenAI model)

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Abstract

Linearized Weyl–Maxwell gravity on the compactified magnetically supported Plebański–Hacyan universe contains the ordinary Einstein–Maxwell modes and additional Weyl modes. We determine whether the additional generic modes are tangent to nonlinear solutions on the fixed compact magnetic bundle. On the closed Cauchy surface $S^1 \times S^2$, the quadratic Taub pairing with a background stabilizer equals the corresponding covariant Lee–Wald moment map. The time-translation pairing is negative definite on the complete real pure-extra generic sector, axial and polar, for every $\ell \geq 2$ and every allowed compact momentum. Consequently no nonzero pure-extra generic tangent admits a fixed-bundle second-order extension.

This obstruction is not stable under mixing with the Einstein sector. The Einstein primary has an indefinite time-translation form. At $\ell = 2, m = 0, k = 0$, an Einstein-minus mode and one extra mode have a unique positive balancing ratio within their declared two-mode span, for which all five background-stabilizer moment maps vanish. We compute its complete quadratic Weyl–Maxwell source. The homogeneous zero-frequency Einstein and extra sources cancel exactly; every remaining homogeneous channel has an explicit correction; and every $\ell = 2, 4$ output is removed by an exact off-shell polar inverse. The four solved action equations and four target Noether identities span all eight ungauged polar equations with determinant (-4), including at zero frequency. Hence this balanced real tangent has an explicit spatially periodic, finite quasiperiodic second-order correction on the same magnetic bundle component.

The result exhibits a sharp nonlinear distinction:

$$\boxed{\text{pure extra generic mode: obstructed,} \qquad \text{balanced Einstein–extra mode: extendible to second order.}}$$

The theorem concerns a formal second-order jet, not an exact family or an all-orders solution. Post-freeze successor certificates now classify the complete finite-harmonic generic $k = 0$ zero cone, one fixed-$|k|$ opposite-momentum cone, and the full exceptional $\ell = 1, k = 0$ all-m pure cone. They sharpen the frontier without changing Theorems A and B below.

1. Introduction

Linearization stability asks whether a solution of the linearized field equations is the first derivative of a family of exact solutions. On a compact Cauchy surface, background symmetries can produce quadratic Taub constraints which are invisible in the linear equations. This phenomenon is classical in Einstein gravity and is naturally expressed through the constraint-adjoint kernel or, equivalently on a closed slice, through the moment map for the background symmetry group [2,10,11]. Higher-derivative gravity has analogous linearization-instability questions [3].

The present problem is unusually clean because the linear phase space is already known exactly. The background

$$M = \mathbb{R}_t \times S^1_L \times S^2$$

is a compactification of the magnetically supported Plebański–Hacyan direct product [4]. With the normalization specified below, it solves both Einstein–Maxwell and Weyl–Maxwell theory. The complete standard Einstein–Maxwell harmonic tangent injects into the Weyl–Maxwell tangent, while the latter contains two extra generic cyclic summands in each parity. Those extra modes are genuine nonradical linear solutions; they are neither Weyl-gauge representatives nor null directions of the Lee–Wald form. The linear phase-space theorem and its conventions are established in the companion paper [5].

The next question is nonlinear:

> Which of the certified extra linear modes satisfy the quadratic integrability constraints, and can an Einstein component cancel their obstruction?

We answer both parts at a sharp, publishable scope.

To our knowledge, no previous Weyl–Maxwell calculation combines a definite all-harmonic obstruction on the new fourth-order primary with an exact Einstein–extra cancellation and a complete second-order correction.

1. **General pure-extra no-go.** Every nonzero real pure-extra generic tangent is obstructed at second order on the fixed magnetic bundle. This holds for both parities, every physical $\ell \geq 2$, every allowed compact momentum, and the completion of finite harmonic sums in the time-translation charge norm defined below.
2. **Explicit balanced extension.** One nonzero real Einstein–extra tangent at $\ell = 2, m = 0, k = 0$ annihilates all five stabilizer moment maps and has a complete explicit second-order correction.

The contrast is the scientific point. The extra sector is not deleted from the linear theory, nor is it generically promoted to a nonlinear branch. Rather, the nonlinear solution locus has a singular tangent cone: a pure extra direction fails the quadratic constraint, while an appropriately balanced mixed direction passes and extends through second order.

This paper does **not** delay that conclusion until the entire indefinite mixed cone is classified. The full common zero locus of $\left(\mu_H, \mu_{P_x}, \mu_{J_1}, \mu_{J_2}, \mu_{J_3}\right)$, including opposite-momentum standing waves and exceptional/global blocks, is a post-freeze successor programme summarized in Section 10.

Main theorems

Let $\mathcal{T}^{\mathrm{gen}}_X$ be the real, locally gauge-reduced, generic extra Weyl–Maxwell tangent on the fixed magnetic bundle, and let u_- and u_e be the axisymmetric axial modes defined in Section 7.

Theorem A (definite pure-extra obstruction). For every nonzero $u \in \mathcal{T}^{\mathrm{gen}}_X$,

$$\mu_H(u) = -\frac{L}{4}\|u\|^2_H < 0.$$

Therefore the second-order Weyl–Maxwell equation has no solution on the fixed magnetic-bundle component P_N.

Theorem B (one balanced second-order jet). Let

$$a_e = \sqrt{\frac{27}{52}(5\sqrt{3} - 6)} > 0.$$

The explicitly defined real $\ell = 2, m = 0, k = 0$ Einstein–extra tangent annihilates all five background-stabilizer moment maps and admits a smooth, real, S^1_L-periodic in space and finite quasiperiodic in time second-order correction on the same bundle component.

Theorem A concerns the complete axial and polar pure-extra generic sector. Theorem B is an existence theorem for one declared mixed tangent. It is not a claim that every common-zero tangent extends.

2. The common background and the fixed magnetic-bundle component

We use signature $(-+++)$ and the actions

$$S_{\mathrm{EM}}[g, A] = \int_M \sqrt{-g}\left[\frac{R - 2\Lambda}{2\kappa} - \frac{1}{4}F_{ab}F^{ab}\right] d^4x,$$

$$S_{\mathrm{WM}}[g, A] = \int_M \sqrt{-g}\left[\frac{\alpha_B}{8}C_{abcd}C^{abcd} - \frac{1}{4}F_{ab}F^{ab}\right] d^4x.$$

The Bach convention is

$$\delta\int \sqrt{-g}\, C^2 = 4\int \sqrt{-g}\, B_{ab}\,\delta g^{ab} + \text{boundary},$$

so the Weyl–Maxwell metric equation is

$$\alpha_B B_{ab} = T_{ab}, \qquad \nabla_a F^{ab} = 0, \qquad dF = 0.$$

The fixture is

$$M = \mathbb{R}_t \times S^1_L \times S^2, \qquad d\bar{s}^2 = -dt^2 + dx^2 + d\Omega_2^2, \qquad \bar{F} = \mathrm{vol}(S^2),$$

with

$$(\kappa, \Lambda, \alpha_B) = (1, \tfrac{1}{2}, 3).$$

It lies on the intersection of the Einstein–Maxwell and Weyl–Maxwell solution loci. This is an incidence relation between two theories, not a Weyl gauge equivalence. The local geometry belongs to the direct-product electrovacua studied by Plebański and Hacyan [4]. Exact wave families on related direct-product universes are known [6,7]; their existence neither implies nor is implied by the complete harmonic integrability statement proved here.

The magnetic field is a connection curvature on a nontrivial compact $U(1)$ bundle $P_N \to S^1 \times S^2$. With the chosen normalization the Chern number is $N = 2$. A connection tangent $a = A - \bar{A}$ is a global one-form on the fixed bundle, but a uniform continuous change of magnetic flux changes the bundle family. We therefore define the nonlinear extension problem on the fixed component P_N.

This distinction is load-bearing. If a formal second-order magnetic coefficient is introduced,

$$N(\epsilon) = 2 + 2\epsilon^2 p,$$

then a smooth family on the same $N = 2$ bundle has $p = 0$. A calculation which removes a Taub source by choosing $p \neq 0$ solves a different, charge-enlarged problem. Electric variation is allowed on fixed P_N, but at the purely magnetic background its linear energy pairing vanishes and it cannot absorb the pure-extra constant-lapse obstruction.

Domain of the theorem

The theorem is made after local

$$\mathrm{Diff} \ltimes \mathcal{G}_{\mathrm{Weyl}} \ltimes \mathcal{G}_{U(1)}$$

reduction, but before quotienting by the five background stabilizers

$$H = \partial_t, \qquad P_x = \partial_x, \qquad J_1, J_2, J_3 \in \mathfrak{so}(3).$$

These stabilizers are retained because their Hamiltonians are the quadratic constraints. Quotienting them before evaluating the moment map would erase the very obstruction under study.

3. Linear input: Einstein and extra primary sectors

Write

$$\lambda = \ell(\ell + 1), \qquad k = \frac{2\pi n}{L}, \quad n \in \mathbb{Z}.$$

For every $\ell \geq 2$, the generic axial and polar Weyl–Maxwell solution modules split into the Einstein q-primary image and two additional p-primary summands. The axial statement is imported from the companion linear paper [5]. Because the polar statement is load-bearing for Theorem A, we record its complete algebraic and current certificate here rather than importing it by name alone.

The extra shell is

$$p(\omega, k, \lambda) = \omega^2 - k^2 - \lambda + \frac{2}{3} = 0,$$

so

$$\omega_e^2 = k^2 + \lambda - \frac{2}{3} > 0.$$

The Einstein shell polynomial is

$$q(\omega, k, \lambda) = (\omega^2 - k^2)^2 - 2\lambda(\omega^2 - k^2) + \lambda(\lambda - 2).$$

Its two positive-frequency branches will be denoted ω_- and ω_+.

3.1 Polar module and current theorem

In the polar gauge slice and coefficient order (A_t, B, C_t, U), the action-normalized reduced Hessian is

$$H_P = \begin{pmatrix} \frac{2k^4+4k^2\lambda+2\lambda^2-3\lambda}{4} & k\omega(k^2+\lambda) & \frac{k^2\lambda+2k^2\omega^2+\lambda^2-\lambda\omega^2+\lambda}{4} & \lambda \\ k\omega(k^2+\lambda) & -\frac{3k^2\lambda-4k^2\omega^2+3\lambda^2-3\lambda\omega^2-2\lambda}{2} & -k\omega(\lambda-\omega^2) & 0 \\ \frac{k^2\lambda+2k^2\omega^2+\lambda^2-\lambda\omega^2+\lambda}{4} & -k\omega(\lambda-\omega^2) & \frac{2\lambda^2-4\lambda\omega^2-3\lambda+2\omega^4}{4} & -\lambda \\ \lambda & 0 & -\lambda & -2\lambda(k^2+\lambda-\omega^2) \end{pmatrix}. \tag{3.1}$$

The normalization follows from the four-dimensional first variation, with row weights $(-1, 2, -1, 2\lambda)$ for the metric 00, metric 01, metric 11, and polar Maxwell-density equations. It is therefore not inferred from formal self-adjointness alone.

Let $I_j(H_P)$ denote the ideal generated by the $j \times j$ minors. Over

$$R^P_{\text{phys}} = \mathbb{Q}[\lambda, k, \lambda^{-1}, (\lambda-2)^{-1}, (3\lambda-2)^{-1}, (3\lambda-4)^{-1}, (5\lambda+6)^{-1}, (9\lambda+2)^{-1}, (9\lambda-2)^{-1}],$$

where neither k, ω, p, nor q is inverted, exact Bézout witnesses give

$$I_1 = (1), \qquad I_2 = (1), \qquad I_3 = (p), \qquad I_4 = (p^2 q). \tag{3.2}$$

The resultant

$$\operatorname{Res}_\omega(p, q) = \frac{4}{81}(9\lambda-2)^2 \tag{3.3}$$

is a unit on every physical fibre. After specialization to a physical (ℓ, n) fibre, $K_{\ell,n}[\omega]$ is a PID and the specialized Fitting ideals (3.2) give the Smith factors $1, 1, p, pq$, including at $k = 0$. No global Smith normal form over the multivariate ring $R^P_{\text{phys}}[\omega]$ is asserted. Fibrewise primary decomposition gives

$$\mathcal{T}^{\text{pol}}_{\text{WM}} \cong K_{\ell,n}[\omega]/(q) \oplus \big(K_{\ell,n}[\omega]/(p)\big)^2. \tag{3.4}$$

Here $K_{\ell,n}$ is any characteristic-zero field containing the specialized coefficients $\lambda = \ell(\ell+1)$ and $k = 2\pi n/L$. The Einstein solution map is injective, is annihilated by q, and has $K_{\ell,n}$-dimension four. Since q is a unit on the two p-primary summands, its image equals the complete q-primary summand.

On the positive extra shell, define

$$D = 3k^2 + 3\lambda - 2, \qquad \widetilde{D} = 6k^2 + 3\lambda - 2.$$

Two explicit extra representatives are

$$e_1 = \begin{pmatrix} 4(3k^2-2) \\ 0 \\ -12(k^2+\lambda) \\ 3\widetilde{D} \end{pmatrix}, \qquad e_2 = \begin{pmatrix} -2kD \\ \omega_e \widetilde{D} \\ -2kD \\ 0 \end{pmatrix}. \tag{3.5}$$

Their direct four-dimensional Lee–Wald Hermitian current Gram matrix is

$$G_X^{\mathrm{pol}} = \begin{pmatrix} 18\lambda(4k^2+\lambda-2)(12k^2+9\lambda-2) & -6k\lambda(3\lambda+2)D \\ -6k\lambda(3\lambda+2)D & \frac{1}{2}\lambda(3\lambda-2)^2D \end{pmatrix}. \tag{3.6}$$

Its first principal minor is positive and

$$\det G_X^{\mathrm{pol}} = 9\lambda^2(\lambda-2)(9\lambda-2)D\widetilde{D}^2 > 0 \tag{3.7}$$

for every physical $\lambda \geq 6$ and allowed k. Direct reduction of the mixed current gives zero modulo (p,q), so this block is orthogonal to the Einstein primary.

Theorem 3.1 (generic polar linear input). On every physical $\ell \geq 2$ compact-momentum fibre, including $k=0$, the locally gauge-reduced polar target has the decomposition (3.4). Its two extra positive-frequency directions are nonradical, orthogonal to the Einstein primary, and have current inertia $(2,0)$.

Equations (3.1)–(3.7), together with the exact Bézout witnesses in the machine-readable supplement, constitute the polar theorem used below. This is an off-shell module and classical current result before the final background-stabilizer quotient; it is not a causal, particle, or quantum claim.

The Einstein and extra primary modules are therefore orthogonal under the direct four-dimensional Lee–Wald current in both parities. Each axial and polar extra Gram matrix G_X^{par} is positive definite. Thus the extra modes are genuine nonnull linear solutions, even though Section 5 proves that they are not fixed-bundle nonlinear tangent directions.

For a positive-frequency coefficient vector c in one branch, parity and spin-ℓ multiplicity block, let W_ℓ be the positive invariant angular Gram form. Reality fixes the negative-frequency and negative-momentum coefficients by conjugation. We use

$$\Phi = \mathrm{Re}(ce^{-i\omega t+ikx})$$

as the real-mode convention.

4. Taub pairings as covariant moment maps

Let $E(\Phi)=0$ denote the Weyl–Maxwell Euler–Lagrange equations and $L = DE|_{\bar{\Phi}}$. We fix the perturbative convention

$$\Phi(\epsilon) = \bar{\Phi} + \epsilon u + \epsilon^2\Phi^{(2)} + O(\epsilon^3). \tag{4.1}$$

Thus a first-order tangent $Lu=0$ extends through second order only if

$$L\Phi^{(2)} = -\frac{1}{2}D^2E|_{\bar{\Phi}}[u,u]. \tag{4.2}$$

Let $\mathcal{C}_\Sigma(\Phi)=0$ be the complete nonlinear Weyl–Maxwell constraint map obtained by the normal and tangential decomposition of the Euler–Lagrange equations on the closed slice $\Sigma = S^1_L \times S^2$. Write its source components as S_A. For a background stabilizer $\widehat{X}$, let $\zeta^A_{\widehat{X}}$ be its element of the constraint-adjoint kernel and define

$$\langle \zeta_{\widehat{X}}, S \rangle_\Sigma := \int_\Sigma \zeta^A_{\widehat{X}} S_A \, d\Sigma_{\bar{h}}, \tag{4.3}$$

where $d\Sigma_{\bar{h}}$ is the background spatial volume measure and the index contraction includes the metric and Maxwell constraint components in the four-dimensional action convention of Section 2. Every adjoint zero mode therefore gives the necessary Taub condition

$$\left\langle \zeta_{\widehat{X}}, \frac{1}{2} D^2 \mathcal{C}_\Sigma|_{\bar{\Phi}}[u,u] \right\rangle_\Sigma = 0. \tag{4.4}$$

The stabilizers are automorphisms of the magnetic bundle, not merely vector fields on the base. Time and circle translations lift with zero vertical part. A sphere rotation is lifted as

$$\widehat{J}_a = (J_a, \chi_a), \qquad \iota_{J_a} \bar{F} + d\chi_a = 0. \tag{4.5}$$

Such a global lift exists because $\mathcal{L}_{J_a} \bar{F} = 0$ and $H^1(S^2) = 0$. It preserves the background connection even though a chosen local monopole potential need not be invariant. Write $\widehat{\mathcal{L}}_{\widehat{X}}$ for the resulting combined diffeomorphism–$U(1)$ action.

Let $\mathcal{T}_{\mathrm{WM}}$ be the smooth harmonic solution space after local $\mathrm{Diff} \ltimes \mathcal{G}_{\mathrm{Weyl}} \ltimes \mathcal{G}_{U(1)}$ reduction and before the five stabilizers are quotiented. On every declared generic primary block, the Lee–Wald form Ω_{WM} is nondegenerate. The closed-slice Hamiltonian identity is the following proposition.

Proposition 4.1 (Taub pairing equals the stabilizer moment map). For every smooth finite-harmonic Jacobi field u and every lifted background stabilizer $\widehat{X}$,

$$\boxed{\left\langle \zeta_{\widehat{X}}, \frac{1}{2} D^2 \mathcal{C}_\Sigma|_{\bar{\Phi}}[u,u] \right\rangle_\Sigma = \mu_{\widehat{X}}(u) = \frac{1}{2} \Omega_{\mathrm{WM}}(u, \widehat{\mathcal{L}}_{\widehat{X}} u).} \tag{4.6}$$

To prove (4.6), differentiate the action Noether-current identity twice, use $E(\bar{\Phi}) = 0$, $Lu = 0$, and $\widehat{\mathcal{L}}_{\widehat{X}} \bar{\Phi} = 0$, and then integrate over Σ. The second variation of the constraint Hamiltonian gives the left side of (4.6), while the variation of the Hamiltonian generator gives the right side. The chosen expansion (4.1) supplies the factor 1/2. Every exact Lee–Wald improvement and bundle-patching corner term integrates to zero on the closed slice. This proves the identity first for smooth finite harmonic sums. Section 5 extends it by continuity in the charge norm. It is the boundaryless specialization of the covariant phase-space construction [8,9].

The exact real-mode moment maps are

$$\mu_H = -\frac{L}{4} \sum \omega^2 c^\dagger (G_{\mathrm{branch}} \otimes W_\ell) c, \tag{4.7}$$

$$\mu_{P_x} = \frac{L}{4} \sum k\omega c^\dagger (G_{\mathrm{branch}} \otimes W_\ell) c, \tag{4.8}$$

$$\mu_{J_a} = \frac{L}{4} \sum \omega c^\dagger (G_{\mathrm{branch}} \otimes W_\ell T_a) c. \tag{4.9}$$

The sums are block diagonal in k, ℓ, parity, and frequency shell. Here

$$T_a = -i\mathcal{L}_{J_a}\big|_{\mathcal{H}_\ell} \tag{4.10}$$

is the Hermitian angular-momentum matrix on the spin-ℓ harmonic space, with Hermiticity taken relative to W_ℓ. Thus $T_3 Y_{\ell m} = m Y_{\ell m}$ and (4.9) is manifestly real. Rotations preserve ℓ; T_3 is diagonal in m, while T_1, T_2 connect only m to $m \pm 1$. Axial–polar and Einstein–extra cross terms vanish. The sign and the factor $1/4$ in (4.7) are fixed by exact agreement with three independent direct tensor calculations: one axial extra block and the axial and polar Einstein-minus fixtures at $\ell = 2, k = 0$.

5. The complete pure-extra obstruction

For a finite real pure-extra harmonic sum, define the time-translation charge norm

$$\|u\|_H^2 := \sum_{\ell,m,n,\mathrm{par}} \omega_e^2\, c_{\ell mn}^\dagger (G_X^{\mathrm{par}} \otimes W_\ell) c_{\ell mn}. \tag{5.1}$$

Let $\mathcal{H}_X$ be the Hilbert completion of finite harmonic sums in this norm. This is the precise infinite-superposition domain used here; no claim about a larger PDE energy space is made.

Theorem 5.1 (pure-extra fixed-bundle no-go). Every nonzero real pure-extra generic tangent $u \in \mathcal{H}_X$, axial or polar, satisfies $\mu_H(u) = -(L/4)\|u\|_H^2 < 0$. If, in addition, u is represented in a regularity class in which the quadratic extension equation (4.2) is defined and the Taub pairing (4.3) is continuous, then u admits no second-order correction on the fixed magnetic bundle P_N.

Proof

On the extra shell, $\omega_e^2 > 0$ for every physical $\lambda \geq 6$. Both extra Gram matrices and the angular form W_ℓ are positive definite. Hence every nonzero block contributes strictly negatively to (4.7):

$$\mu_H(u) = -\frac{L}{4} \sum_{\text{extra blocks}} \omega_e^2 c^\dagger (G_X^{\mathrm{par}} \otimes W_\ell) c < 0.$$

Thus $\mu_H(u) = -(L/4)\|u\|_H^2$. Proposition 4.1 and the moment map extend continuously from finite sums to $\mathcal{H}_X$, and strict negativity holds for every nonzero vector in the completion. Equation (4.4) is therefore violated for the constant-lapse adjoint class.

The obstruction cannot be absorbed inside the declared domain. Continuous magnetic variation changes $c_1(P_N)$, while electric variation has zero linear pairing with H at the purely magnetic background. Thus (4.2) has no fixed-bundle solution. □

What the theorem does not say

The theorem does not remove the extra linear modes. Their Lee–Wald Gram matrices are nondegenerate, so they remain genuine classical linear solutions. The conclusion is instead that the exact fixed-bundle solution locus is singular at the background: its formal tangent space is larger than its second-order tangent cone.

Nor is this a quantum ghost statement. No positive-frequency Hilbert space, BRST-compatible Hadamard state, or Lorentzian causal quantum construction is used.

6. Why an Einstein component can cancel the obstruction

The Einstein q-primary contribution to μ_H is indefinite. In each parity its two master branches contribute opposite signs in the target current convention. Consequently a negative extra contribution can be cancelled by an Einstein-minus component without invoking a mixed Einstein–extra current entry. The cancellation is additive between two orthogonal diagonal primary blocks.

There is nevertheless a useful same-momentum restriction.

> **Proposition 6.1 (one travelling block at nonzero momentum).** In a single fixed $k \neq 0$ travelling block, simultaneous vanishing of μ_H and μ_{P_x} forces all Einstein-plus, Einstein-minus, and extra occupations to vanish.

Indeed, the three shell frequencies obey

$$\omega_- < \omega_e < \omega_+.$$

After eliminating the Einstein-minus occupation from the two scalar moment maps, one obtains

$$\omega_+(\omega_+ - \omega_-)A_+ + \omega_e(\omega_e - \omega_-)A_e = 0,$$

where $A_+, A_e \geq 0$ are Gram-normalized occupations. Both coefficients are strictly positive, so $A_+ = A_e = 0$, and the remaining occupation then vanishes as well. This proposition does not cover cancellations between distinct momenta, such as standing-wave combinations.

At $k = 0$, the momentum constraint vanishes automatically and a nontrivial balance becomes possible.

7. The minimal balanced tangent

Fix the axial $\ell = 2, m = 0, k = 0$ sector. In the coefficient order (H_t, H_x, Q_t, Q_x), choose the Einstein-minus representative

$$u_- = (0, -2, 0, 2\sqrt{3}), \qquad \omega_-^2 = 6 - 2\sqrt{3}, \tag{7.1}$$

and the second extra representative

$$u_e = (0, -\tfrac{2}{3}, 0, 6), \qquad \omega_e^2 = \frac{16}{3}. \tag{7.2}$$

For unit real cosine amplitude, their constant-lapse Taub coefficients are

$$\tau_- = \frac{48}{5}(-6 + 5\sqrt{3}) > 0, \qquad \tau_e = -\frac{832}{45} < 0. \tag{7.3}$$

Fix the positive real balancing amplitude

$$a_e := \sqrt{\frac{\tau_-}{-\tau_e}} = \sqrt{\frac{27}{52}(5\sqrt{3} - 6)} > 0. \tag{7.4}$$

Then

$$\Phi^{(1)} = \operatorname{Re}(u_- e^{-i\omega_- t}) + a_e \operatorname{Re}(u_e e^{-i\omega_e t}) \tag{7.5}$$

is nonzero and satisfies

$$\mu_H = \mu_{P_x} = \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = 0. \tag{7.6}$$

Here $\mu_H = 0$ follows from (7.3)–(7.4), $\mu_{P_x} = 0$ from $k = 0$, and the rotational expectations vanish for the separate axisymmetric $m = 0$ states. Because the Einstein and extra primaries are symplectically orthogonal, (7.6) is not produced by an interference term.

The declared theorem concerns this positive-real relative phase. Moreover,

$$\left(\frac{\omega_e}{\omega_-}\right)^2 = \frac{4}{3} + \frac{4\sqrt{3}}{9} \notin \mathbb{Q}, \tag{7.7}$$

so the mixed tangent is not periodic in time. It is smooth and periodic in $x \in S^1_L$, and its time dependence is a finite quasiperiodic sum.

Vanishing Taub charges is necessary but not sufficient for extension. A nonzero quadratic source can still have a component in another adjoint cokernel or lie on a resonant output shell. We therefore solve the complete second-order equation directly.

8. Complete second-order extension of the balanced tangent

Theorem 8.1 (balanced Einstein–extra second-order extension). The real tangent (7.5), with the positive real amplitude (7.4), admits a smooth, real correction $\Phi^{(2)}$ which is S^1_L-periodic in space and a finite quasiperiodic sum in time on the same fixed magnetic-bundle component. It satisfies

$$L_{\mathrm{WM}}\Phi^{(2)} = -\frac{1}{2} D^2 E_{\mathrm{WM}}[\Phi^{(1)}, \Phi^{(1)}]. \tag{8.1}$$

8.1 Selection rules and real-mode factors

The product of two axial $\ell = 2, m = 0$ harmonics is polar and contains only

$$\ell_{\mathrm{out}} = 0, 2, 4.$$

The time dependence produces the five channel types

Channel	Output frequency
Einstein self-sum	$2\omega_-$
extra self-sum	$2\omega_e$
cross-sum	$\omega_e + \omega_-$
cross-difference	$\omega_e - \omega_-$
conjugate self-products	(0)

For $\Phi = \operatorname{Re} z = (z + \bar{z})/2$ and symmetric Hessian B,

$$\frac{1}{2}B(\Phi,\Phi)=\frac{1}{8}B(z,z)+\frac{1}{4}B(z,\bar{z})+\frac{1}{8}B(\bar{z},\bar{z}).$$

Thus self-sums carry factor 1/8, while self-zero, cross-sum, and cross-difference terms carry factor 1/4. These factors are replayed symbolically in the certificate rather than inserted as conventions after the calculation.

8.2 The exceptional homogeneous channel

In homogeneous coefficient order (C,K,U), the directly computed linear operator on rows $(E_{00},E_{11},E_{22},M_1)$ is

$$L_0(\Omega)=\begin{pmatrix} 0 & 0 & 0 \\ -\Omega^4/2 & \Omega^4/2 & 0 \\ \Omega^4/4 & -\Omega^4/4 & 0 \\ 0 & 0 & \Omega^2 \end{pmatrix}. \tag{8.2}$$

At zero frequency the separate real self-products are obstructed:

$$S_-^{(0)}=\frac{-6+5\sqrt{3}}{5}\begin{pmatrix}48\\0\\24\\0\end{pmatrix},\qquad S_e^{(0)}=-\frac{-6+5\sqrt{3}}{5}\begin{pmatrix}48\\0\\24\\0\end{pmatrix}. \tag{8.3}$$

The amplitude (7.4) is already included in $S_e^{(0)}$. Therefore

$$S_-^{(0)}+S_e^{(0)}=0. \tag{8.4}$$

This cancellation is the nonlinear heart of the construction. Each pure component fails the homogeneous constraint, while the balanced combination removes it exactly.

Every nonzero-frequency homogeneous source has

$$S_{00}=S_{M_1}=0,\qquad S_{11}+2S_{22}=0.$$

It is removed explicitly by

$$(C,K,U)=\left(\frac{2S_{11}}{\Omega^4},0,0\right). \tag{8.5}$$

Set

$$C_E=-\frac{3(-36+17\sqrt{3})}{20(-3+\sqrt{3})^2},\qquad C_X=\frac{9(-6+5\sqrt{3})}{1280}.$$

With the common certificate prefix `homogeneous_channels.`, the homogeneous channel ledger is

pair	Ω	correction (C, K, U)	key suffix	R
balanced zero	0	$(0, 0, 0)$	`combined_zero`	0^4
Einstein self	$2\omega_-$	$(C_E, 0, 0)$	`Einstein_self_sum`	0^4
extra self	$2\omega_e$	$(C_X, 0, 0)$	`extra_self_sum`	0^4
cross sum	$\omega_e + \omega_-$	$(2S_{11}/\Omega^4, 0, 0)$	`cross_sum`	0^4
cross difference	$\omega_e - \omega_-$	$(2S_{11}/\Omega^4, 0, 0)$	`cross_difference`	0^4

The separate zero-frequency Einstein and extra rows are the two nonzero vectors in (8.3); only their balanced sum belongs to the solvable ledger.

Hence no electric-charge or Wilson-line correction is hidden in the homogeneous solution.

8.3 The $\ell = 2, 4$ polar outputs

Let $H_P(\lambda, k, \Omega)$ be the exact action-normalized polar Hessian in coordinates (A_t, B, C_t, U). Its determinant is

$$\det H_P = \frac{9}{16}\lambda^3(\lambda - 2)p(\Omega, k, \lambda)^2 q(\Omega, k, \lambda). \tag{8.6}$$

For every output frequency in the table and for $\lambda = 6, 20$, the exact algebraic preflight proves

$$p \neq 0, \qquad q \neq 0.$$

Thus each polar source $S_{\ell,\Omega}$ has the explicit correction

$$\Phi^{(2)}_{\ell,\Omega} = -H_P(\lambda, 0, \Omega)^{-1} S_{\ell,\Omega}, \qquad \ell = 2, 4, \tag{8.7}$$

and every stored four-row remainder is identically zero. The complete finite channel ledger is:

With the common certificate prefix `generic_polar_channels.` and $E =$ Einstein, $X =$ extra, the ledger is below. Every listed four-row remainder is 0^4.

ℓ	pair	Ω	$\operatorname{sgn}(p, q)$	key suffix
2	balanced zero	0	$(-, +)$	`2.combined_zero`
2	EE	$2\omega_-$	$(+, +)$	`2.Einstein_self_sum`
2	XX	$2\omega_e$	$(+, +)$	`2.extra_self_sum`
2	EX	$\omega_e + \omega_-$	$(+, +)$	`2.cross_sum`
2	$E\bar{X}$	$\omega_e - \omega_-$	$(-, +)$	`2.cross_difference`
4	balanced zero	0	$(-, +)$	`4.combined_zero`
4	EE	$2\omega_-$	$(-, +)$	`4.Einstein_self_sum`
4	XX	$2\omega_e$	$(+, -)$	`4.extra_self_sum`
4	EX	$\omega_e + \omega_-$	$(-, -)$	`4.cross_sum`
4	$E\bar{X}$	$\omega_e - \omega_-$	$(-, +)$	`4.cross_difference`

The signs are exact ordered-algebraic-field results, not floating-point tests. In particular, every listed p and q is nonzero. The separate Einstein and extra zero-frequency sources are intentionally absent from the ledger: neither is solvable alone; their weighted sum is the `combined_zero` row.

For visibility, two nontrivial corrections in the order (A_t, B, C_t, U) are

$$\Phi^{(2)}_{2,2\omega_e} = (-6+5\sqrt{3}) \begin{pmatrix} 45723/91364 \\ 0 \\ 9873/91364 \\ -2097/22841 \end{pmatrix}, \tag{8.8}$$

$$\Phi^{(2)}_{4,2\omega_e} = (-6+5\sqrt{3}) \begin{pmatrix} 2097/19565 \\ 0 \\ 6183/19565 \\ -351/1505 \end{pmatrix}. \tag{8.9}$$

Direct multiplication by H_P returns minus the stored source in all four rows for each vector. The larger cross-channel radicals remain in the machine-readable supplement, where their source, correction, and zero remainder appear under the keys displayed in the ledger.

8.4 Completion of the dependent tensor equations

Solving four action rows is sufficient only if the remaining ungauged tensor equations are proved to follow. Use target equation order

$$(A, B, C, h_t, h_x, K, G, U).$$

At $k = 0$, the four certified target Noether identities are the rows of

$$N_0 = \begin{pmatrix} 2i\Omega & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & i\Omega & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & i\Omega & 0 & -\lambda & 2 & -1 \\ -2 & 0 & 2 & 0 & 0 & 2 & 0 & 0 \end{pmatrix}. \tag{8.10}$$

Let $S_{A,B,C,U}$ select equations A, B, C, U. Exact calculation gives

$$\det \begin{pmatrix} S_{A,B,C,U} \\ N_0 \end{pmatrix} = -4. \tag{8.11}$$

The determinant is a nonzero constant, independent of Ω and λ. Therefore the four solved equations and four identities span all eight equations, including at $\Omega = 0$.

This applies to the quadratic source without assuming that $\Phi^{(1)}$ integrates to an exact family. If the nonlinear Noether identity is $N(\Phi)E(\Phi) = 0$, then expansion about an on-shell background with an on-shell first-order tangent gives

$$N^{(0)}E^{(2)} + N^{(1)}E^{(1)} + N^{(2)}E^{(0)} = 0 \quad \Longrightarrow \quad N^{(0)}E^{(2)} = 0. \tag{8.12}$$

Thus the dependent quadratic rows obey precisely the background identities used in (8.11).

8.5 Reality and fixed-magnetic-bundle completion

Define $\Phi^{(2)}$ as the finite sum of (8.5), (8.7), their complex conjugates, and the real zero-frequency corrections. Then $\Phi^{(2)}$ is real, S^1_L-periodic in space, and finite quasiperiodic in time. No independent homogeneous solution is added. The general problem fixes only the magnetic-bundle component P_N; the constructed correction additionally leaves the homogeneous electric and Wilson-line coordinates unchanged:

- the magnetic Chern-class shift is zero;
- every $\ell = 2, 4$ Maxwell correction integrates to zero on S^2;
- the homogeneous Maxwell coefficient U is zero in every channel;
- there is no stationary electric-charge or Wilson-line zero-mode shift.

Equations (8.2)–(8.12) prove (8.1). □

9. Interpretation: a singular nonlinear solution cone

The two theorems fit together without tension.

linear direction	quadratic Taub data	second-order status
nonzero pure extra generic	$\mu_H < 0$	obstructed
balanced $u_- + a_e u_e$	$\mu_H = \mu_{P_x} = \mu_{J_i} = 0$	explicitly extendible

Schematically, the geometry is

$$\begin{array}{ccccc} \mathcal{T}_X \setminus \{0\} & \subset & \mathcal{T}_{\mathrm{lin}} = \mathcal{T}_E \oplus \mathcal{T}_X & \supset & \mathbb{R}_{>0}(u_- + a_e u_e) \\ \mu_H < 0 & & \mu_H \text{ indefinite} & & \mu_H = \mu_{P_x} = \mu_{J_i} = 0 \\ \Downarrow & & & & \Downarrow \\ \text{outside the second-order cone} & & & & \text{certified ray inside the second-order cone.} \end{array} \tag{9.1}$$

This is not a contradiction between the linear and nonlinear analyses. The linearized equations compute the formal tangent space $\ker L/\text{local gauge}$. The quadratic Taub map cuts out the second-order tangent cone inside it. Positive definiteness on the pure-extra subspace excludes every nonzero ray in that subspace. Indefiniteness after adding the Einstein sector creates null directions of the quadratic map, one of which is shown here to satisfy the entire second-order equation.

The result also separates two notions often conflated in discussions of higher-derivative gravity.

1. **A linear mode can be physical at linear order.** The extra modes are nonradical under the classical Lee–Wald current.
2. **A linear mode need not be nonlinearly integrable.** Pure extra modes violate a compact global constraint.

Nothing here identifies a quantum state norm or proves a quantum ghost. The obstruction is classical and global on the closed spatial slice.

The balanced result is stronger than mere Taub cancellation. A nonzero quadratic defect is not a no-go unless it has a nonremovable cokernel component, and vanishing moment maps do not by themselves prove extension. Here the complete source is solved, channel by channel, and the dependent equations are closed by a constant-determinant Noether argument.

10. Why the full nonlinear cone is the next theorem

Post-freeze enlargement of the second-order cone

The frozen theorem proves a general no-go on one large linear subspace and an existence result on one nontrivial mixed ray. Subsequent certificates answer three parts of the larger classification problem without altering that theorem boundary.

The relevant common zero locus is

$$\mathcal{Z}_2 = \{u : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}$$

and the landed results are:

1. every finite-harmonic generic $k = 0$ tangent on this zero locus has a smooth, spatially periodic, finite-quasiperiodic second-order correction;
2. for one generic $\ell \geq 2$ and fixed nonzero $|k|$, the complete both-momentum-sign, all-m, both-parity zero cone has a smooth-global second-order correction, allowing secular time dependence at resonances;
3. every nonzero exceptional $\ell = 1, k = 0$ axial-plus-polar all-m pure tangent is obstructed: distinct-m interference cannot cancel its positive-positive resonance.

The remaining cone problem includes multiple $|k|$ fibres, infinite-mode Sobolev completion, homogeneous/twist/Wilson-line/charge mixtures, and the quadratic-source disposition of their surviving strata. None of the post-freeze steps is needed to validate Theorems 5.1 and 8.1, and no scoped second-order extension is extrapolated into an all-orders closure theorem.

A further exact spectrum census removes one possible loophole. The complete homogeneous $\ell = 0, k = 0$ physical quotient is empty at every nonzero frequency, so there is no hidden homogeneous fourth-order oscillator. In the full $k = 0$ positive-sum spectrum, the only new resonance not already covered by the exceptional all-m theorem is a generalized homogeneous/twist zero-mode multiplied by an $\ell = 2$ extra-primary mode at $\omega^2 = 16/3$. Its bilinear source coefficient is now the sharp next gate; frequency differences and opposite nonzero momenta remain separate.

11. Scope boundary

The established statements are:

- the covariant moment-map/Taub identity on the declared compact generic solution space;
- the complete pure-extra generic fixed-bundle second-order no-go;
- the trivial common H, P_x zero locus in one nonzero-k travelling block;
- one all-stabilizer-zero Einstein–extra $k = 0$ tangent;
- one complete explicit second-order correction for that tangent.

Post-freeze certified successors additionally establish:

- complete finite-harmonic generic $k = 0$ second-order extension on the full stabilizer-zero cone;
- smooth-global second-order extension on one fixed-$|k|$ opposite-momentum cone, with boundedness not claimed;
- the full exceptional $\ell = 1, k = 0$ all-m pure-sector resonance no-go.

The following remain open:

- mixed cones involving multiple $|k|$ fibres or infinite-mode completion;
- general mixed second-order closure;

- integration of the certified jet to an exact or all-orders family;
- homogeneous, twist, Wilson-line, charge, and other generalized global mixtures not covered by the all-m exceptional theorem;
- a final background-stabilizer quotient or relational observable;
- Lorentzian causal propagation, asymptotic scattering, particles, quantization, ghosts, and unitarity.

The paper therefore carries the dependency labels `LOCAL-ALGEBRAIC` and `REDUCED-MODE`, not `LORENTZIAN-CAUSAL`.

12. Computational proof and reproducibility

All ranks, signs, polynomial nonvanishing tests, amplitudes, source projections, and operator remainders are computed in exact rational or algebraic arithmetic. The principal certificates are:

Result	Certificate
fixed-bundle domain and Taub descent	`bridge/certificates/compact_harmonic_domain_taub_desc`
direct generic axial extra current	`bridge/certificates/einstein_maxwell_weyl_axial_lee_w`
physical-ring generic polar module	`bridge/certificates/einstein_maxwell_weyl_polar_phys`
direct generic polar extra current	`bridge/certificates/einstein_maxwell_weyl_polar_lee_w`
direct axial/polar fixture Taub matrices	`bridge/certificates/einstein_maxwell_weyl_hermitian_a`
generic moment-map bridge and pure-extra no-go	`bridge/certificates/einstein_maxwell_weyl_moment_map_`
mixed zero-locus fixture and off-shell preflight	`bridge/certificates/einstein_maxwell_weyl_mixed_momer`
complete balanced correction	`bridge/certificates/einstein_maxwell_weyl_balanced_el`
complete finite-harmonic generic $k = 0$ cone	`bridge/certificates/einstein_maxwell_weyl_finite_harn`
fixed-$\lvert k \rvert$ opposite-momentum cone	`bridge/certificates/einstein_maxwell_weyl_opposite_mo`
exceptional $\ell = 1$ all-m pure no-go	`bridge/certificates/einstein_maxwell_weyl_exceptional`
homogeneous nonzero-frequency quotient	`bridge/certificates/einstein_maxwell_weyl_homogeneous`
exceptional positive-sum resonance census	`bridge/certificates/einstein_maxwell_weyl_exceptional`

Fast verification:

```
python3 bridge/einstein_sector/verify_charge_fibre_paper_claim_map.py
python3 bridge/einstein_sector/verify_einstein_maxwell_weyl_polar_physical_completion.py
python3 bridge/einstein_sector/verify_einstein_maxwell_weyl_polar_lee_wald_gate.py
python3 -m bridge.einstein_sector.einstein_maxwell_weyl_moment_map_taub_bridge \
  --verify bridge/certificates/einstein_maxwell_weyl_moment_map_taub_bridge.json
python3 -m bridge.einstein_sector.verify_einstein_maxwell_weyl_moment_map_taub_bridge
python3 -m bridge.einstein_sector.einstein_maxwell_weyl_mixed_moment_map_zero_locus \
  --verify bridge/certificates/einstein_maxwell_weyl_mixed_moment_map_zero_locus.json
python3 -m bridge.einstein_sector.verify_einstein_maxwell_weyl_mixed_moment_map_zero_locus
python3 -m bridge.einstein_sector.einstein_maxwell_weyl_balanced_ell0_second_order \
  --verify bridge/certificates/einstein_maxwell_weyl_balanced_ell0_second_order.json
python3 -m bridge.einstein_sector.verify_einstein_maxwell_weyl_balanced_ell0_second_order
python3 -m unittest \
  bridge.einstein_sector.tests.test_einstein_maxwell_weyl_polar_physical_completion \
  bridge.einstein_sector.tests.test_einstein_maxwell_weyl_polar_lee_wald_gate \
  bridge.einstein_sector.tests.test_einstein_maxwell_weyl_moment_map_taub_bridge \
  bridge.einstein_sector.tests.test_einstein_maxwell_weyl_mixed_moment_map_zero_locus \
  bridge.einstein_sector.tests.test_einstein_maxwell_weyl_balanced_ell0_second_order
```

The final exhaustive balanced source-and-channel regeneration passed in `468.66` seconds:

```
python3 -m bridge.einstein_sector.einstein_maxwell_weyl_balanced_ell0_second_order \
  --verify-exhaustive \
  bridge/certificates/einstein_maxwell_weyl_balanced_ell0_second_order.json
```

The committed-certificate verifier separately reconstructs the Noether completion determinant, checks all imported content hashes, replays the real-channel factors, and verifies the fixed-magnetic-bundle and homogeneous charge-coordinate flags. The largest nested-radical cross-channel equations are replayed by the exhaustive rail.

Model authorship and human accountability

GPT-5.6.sol, an OpenAI model, contributed the research programme, mathematical direction, derivations, symbolic-code generation and debugging, claim auditing, literature organization, and manuscript. The project was commissioned and directed by Asger Alstrup Palm, who initiated the questions, served as the non-technical orchestrator and corresponding human contact, but claims no technical contribution. Circulation or submission remains conditional on a documented human verification of the mathematical claims, proof boundaries, source citations, and final text.

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Frozen claim flag: `PURE_EXTRA_GENERIC_NO_GO_AND_ONE_BALANCED_MIXED_SECOND_ORDER_EXTENSION_CERTIFIED`

Next theorem: compute the unique surviving positive-sum bilinear between a homogeneous/twist global zero-mode and an $\ell = 2$ extra primary; then dispose frequency-difference, multi-$|k|$, infinite-mode, charge/Wilson-line, and higher-order gates.